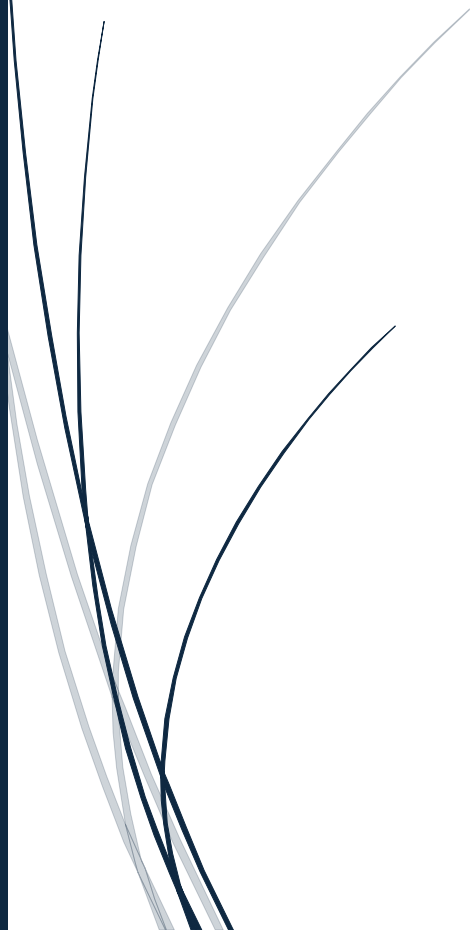




4/2/2024

Informatics 3 Year Project

Deliverable 1: Requirements and Analysis



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1. Background, Problem Statement & Proposed Solution

Introduction

AutoWorx Barnyard is a mechanical car workshop, where precision meets proficiency, and the vehicle's needs take priority. The workshop is dedicated to offering comprehensive automotive services, ensuring vehicles operate smoothly and safely on the road.

In this workshop, skilled mechanics undertake various tasks, ranging from routine maintenance and diagnostics to complex repairs and upgrades. With years of experience and expertise in automotive engineering, the team is committed to delivering exceptional service with diligence and efficiency (Barnyard).

According to (Corrin, 2023) Most workshops without software encounter various challenges. After, identifying these challenges in AutoWorx Barnyard, we approached the workshop management to develop a management system that the workshop can use to combat these challenges. By implementing this management system, the workshop aims to elevate customer experience, providing peace of mind knowing that vehicles are in capable hands.

Below we delve deeper into the dynamic world of mechanic car workshops, exploring their challenges, the services they provide, the systems currently in place, and the proposed solution.

Company / Industry background

According to (JMC Automotive Equipment, 2022), Car workshops are critical to the automobile industry. They provide the expertise and resources required to repair and maintain vehicles. Without them, vehicle owners would be unable to keep their vehicles in optimal condition. This industry plays a crucial role in keeping vehicles roadworthy.

Autoworx Barnyard:

AutoWorx Barnyard, based in Melville, Johannesburg, is one of these workshops. AutoWorx Barnyard is an independent workshop, specializing in the repair and servicing of vehicles. AutoWorx Barnyard offers a range of services to meet the needs of customers. Their top services include suspension checks, gearbox maintenance, repairs, electrical diagnostics, and repairs, as well as tire services, wheel alignment, and a 50 Point Car Service Check which is the overall diagnostics of the vehicle. On 50 Point Check Every aspect of the car is assessed to identify faults that might be disclosed. Below is the breakdown of the service:

Lights:

- Low Beam
- High Beam
- Brake Lights
- Back Up Lights
- Left Turn Signal
- Right Turn Signal
- Taillights
- Emergency Flasher

Guage:

- Fuel
- Oil Pressure
- Volt/Amps
- Temperature

Exterior:

- Tire Thread (1/16")
- Tire Air Pressure
- Windshield Condition
- Body Damage/Loose Parts
- Mirrors
- Wiper Blades

According to the workshop (Barnyard) the following is not required for inspection but is recommended:

Fluid Levels:

- Oil
- Coolant
- Brake
- Power Steering
- No Leaks
- Belts not frayed/cracked/loose
- Battery Connection clean/tight
- Hoses (no cracks or leaks)
- Steering
- Shock Absorbers/Struts

Industry:

An automobile repair workshop is a place where skilled mechanics repair vehicles. A wide range of tasks, requiring a diverse set of skills, competence, aptitude, and knowledge is conducted according to (Leeming, 1974). They execute a variety of tasks, including oil changes, brake replacement, and engine components replacement. (Buerkle Body Shop, 2014) Workshops oversee inspecting, maintaining, and repairing cars to ensure their safety.

Current system(s) and/or procedures followed:

Autoworx Barnyard:

The workshop engages with clients through various channels, namely walk-ins, email, telephone, WhatsApp, the website, and callouts. Quotes and invoices are generated manually using a Microsoft Word template. Mechanics' tasks are delegated in person each morning, with a job card utilized for the 50-point car service check. Progress reports are usually sent to clients via WhatsApp, and client records are stored as a collection of Word document invoices on the manager's computer.

Industry:

According to (Schmidt & Withopf, 2014) The current car workshop systems often involve various aspects such as mechanical diagnostics, service management, vehicle repair, and automotive maintenance. These systems require efficient estimation of time-variant acoustic feedback paths in in-car communication systems.

Problem Statement:

The current AutoWorx Barnyard's customer communication and inquiries, invoice & quotation generation, task coordination, and process documentation rely on outdated methods which include the use of telephone & WhatsApp for communication, Microsoft Word Invoice & Quotation Template, task assignment by mouth resulting in a lack of systematic customer relationship management, task delegation, and process documentation hindering productivity, customer engagement, and knowledge sharing.

Proposed Solution:

We will develop a workshop management system that prioritizes customer engagement, seamless communication, and efficient service estimation costs. This system aims to improve operations by enhancing transparency, delegating mechanics' tasks, automating the quoting and invoice generation process, maintaining comprehensive client service history, and promoting skill sharing among team members to ensure consistent service quality.

Some of the benefits of this system are:

Enhanced Customer Experience: The customer portal provides transparency and real-time updates on their vehicle's status.

Efficient Quote Generation: Automated quotes streamline the estimation process, improving accuracy and transparency.

Effective Communication: The communication hub fosters direct interaction between customers and the workshop, addressing inquiries and collecting valuable feedback.

Improved Customer Retention Rate: By offering personalized notifications to clients for future services, the chances of clients returning increases.

Improved knowledge sharing: A mechanism for standardizing procedures and best practices will improve knowledge sharing among mechanics.

Simplified Business Reports: This will help to identify the areas for improvement, measure profitability, and make data-driven decisions.

2. Requirements Extraction

Functional Requirements

1. User Management:

a) Registration:

- Customers should be able to register accounts by providing required information such as name, email, and password.
- The provided details should be validated to prevent duplicates.
- Manager should be able to register customer, manager, and mechanic.

b) Login:

- Registered users should be able to log in to their accounts using their credentials.
- Users' credentials should be authenticated before granting them access to the appropriate dashboard.
- All passwords should be hashed to protect user credentials.

c) Update Profile:

- Users should have the ability to manage and modify their profile information.

d) Logout:

- Users should be able to log out of the system or be logged out automatically if their session times out for security purposes.

e) Delete Account:

- A Client should be able to delete their account.
- The Manager should have the ability to delete the Mechanic account, Client Account, and their account.
- Mechanic should not have the ability to delete their account.

2. Service Request Management:

a) Request for a Quote:

- Customers should be able to request for a quote for servicing a car.

b) Make service request:

- Customers should be able to navigate through a list of mechanical services offered by the workshop and select the service(s) they need.
- If a service is not listed the customer provides options for customers to describe issues or provide additional information on their service request.

c) Manage Requests:

- The Manager should be able to view the list of service requests, accept or decline requests, and produce a quotation along with a list of available dates and time slots for an appointment.
- The customer should have an option to request a service once they receive a quotation.

d) View Quote:

- Customers should be able to view generated quotes for requested services, including a breakdown of costs.
- Quotes should be generated using a template, detailing labour, parts, taxes, and any additional fees.
- Allow customers to request a service with the quotes provided.

e) Send an Invoice:

- After a service has been requested the manager should be able to send an invoice to the customer.

f) Service History:

- Record details of past repairs performed on each registered vehicle, including the nature of repairs, parts replaced, and associated costs.
- A customer should be able to view the repair history of their vehicles.
- The manager should be able to view the history of vehicles serviced.
- Display the service logs including past repairs, maintenance tasks, service dates, and technician notes.

- Customer should be able to see all their service history and request for a service again. The prices are subject to change when a new invoice is generated.

3. Task Management:

a) View Task:

- Mechanics should be able to view assigned tasks in their dashboard, including details such as service requirements and vehicle information.

b) Update Task:

- Mechanics should be able to update task statuses, marking tasks as in progress, completed, or requiring further attention.
- Provide options for mechanics to add comments or notes to task updates.
- Task updates should be reflected in real-time across the system.

c) Assign Task:

- Managers should be able to assign tasks to mechanics.
- Provide options for managers to specify task details and deadlines.

4. Performance Monitoring:

a) Generate Report:

- Manager should be able to generate reports on key metrics such as task completion rates, service turnaround times, and customer feedback scores.

b) View Reports

- Managers should have access to performance analytics and reports to evaluate workshop efficiency, resource utilization, and customer satisfaction.

5. Appointment Management:

a) Scheduling Appointments:

- Customers should be able to schedule appointments on the platform, specifying preferred dates, times, and service requirements.

- Ensure that appointment scheduling does not conflict with existing appointments.
- Managers should be able to view appointments customers requested.
- Managers should be able to update appointment dates.

6. Skills Sharing:

a) Add to skills sharing:

- Managers should be able to add to skills sharing for every vehicle being serviced. This is done by selecting an existing vehicle or creating a new form for a new vehicle and adding which part was being serviced and how it was serviced.

b) View skills sharing:

- Mechanics and should be able to view skills for any car they wish to by searching for the car name and selecting a model.

Non-Functional Requirements

For Admin:

1. Security

- Use of Data Encryption: Sensitive data such as payment information, customer information, mechanic information, and manager information are encrypted to prevent unauthorized access. due to the POPI Act of 2013, the workshop is responsible for protecting the personal information of its clients. This system will ensure System security through Secure User Authentication and Role-based Access control.
- Role-Based Access Control: Each manager has a specified level of access to prevent unauthorized managers from changing manager, mechanic, and customer information.

2. System Reliability

- Availability: The website the manager will be using should have an uptime of 100% except when the system is being maintained. Managers should be able to access the website system at any time without disrupting system operations.

3. Scalability

- Managers Access: Allow a maximum of at least 100 managers to concurrently use the website system without affecting the system's high performance. This ensures that the system can accommodate increasing user demand as the business grows.

4. Performance

- Website should have a maximum loading time of 5 seconds when the server is operating at normal conditions. This allows managers to efficiently navigate and manage tasks without experiencing delays, this enhances the productivity of the workshop.

5. Usability

- The manager should be able to access the website using major web browsers (Chrome, Edge, Safari, Firefox) and accessible on desktop and tablet devices to ensure that they can access the system across different platforms which will improve usability and accessibility.

For Customer:

1. Security

- Customer accounts must use strong authentication mechanisms, this means that password hashing will be used. This ensures there is security for customer's data and service history.

2. Performance

- Application should have a maximum loading time of 5 seconds when the server is operating at normal conditions. This allows customers to efficiently navigate and view services without experiencing delays, which enhances the user experience.

3. Device Compatibility

- The app should be compatible with a wide range of Android versions running on a mobile device starting from at least Android 5.0.

4. Reliability

- The Customer should have uptime of 100% except when the system is being maintained. Customers should have access to services and information, avoiding disruptions to customer interactions and transactions.

5. Personalization

- The application supports personalized recommendations based on service history, and other past service interactions (Quote requests, Invoices).

6. Notification System

- Application will provide notifications to car serving progress to customers, quotes that are sent by the manager, and invoices.

For Mechanics:

1. Security

- Mechanic accounts must use strong authentication mechanisms, this means that password hashing will be used. This ensures there is security for mechanic data and service history.

2. Performance

- Application should have a maximum loading time of 5 seconds when the server operates at normal conditions. This allows mechanics to efficiently update service progress without experiencing delays, which enhances the service operations.

3. Real-Time Updates

- Any service progress update made by the mechanic must be reflected in real-time on the manager's website and customer app. This allows transparent communication between mechanics, managers, and customers, enhancing service quality.

3. Identification and Prioritisation of Use Cases

1. User Management:

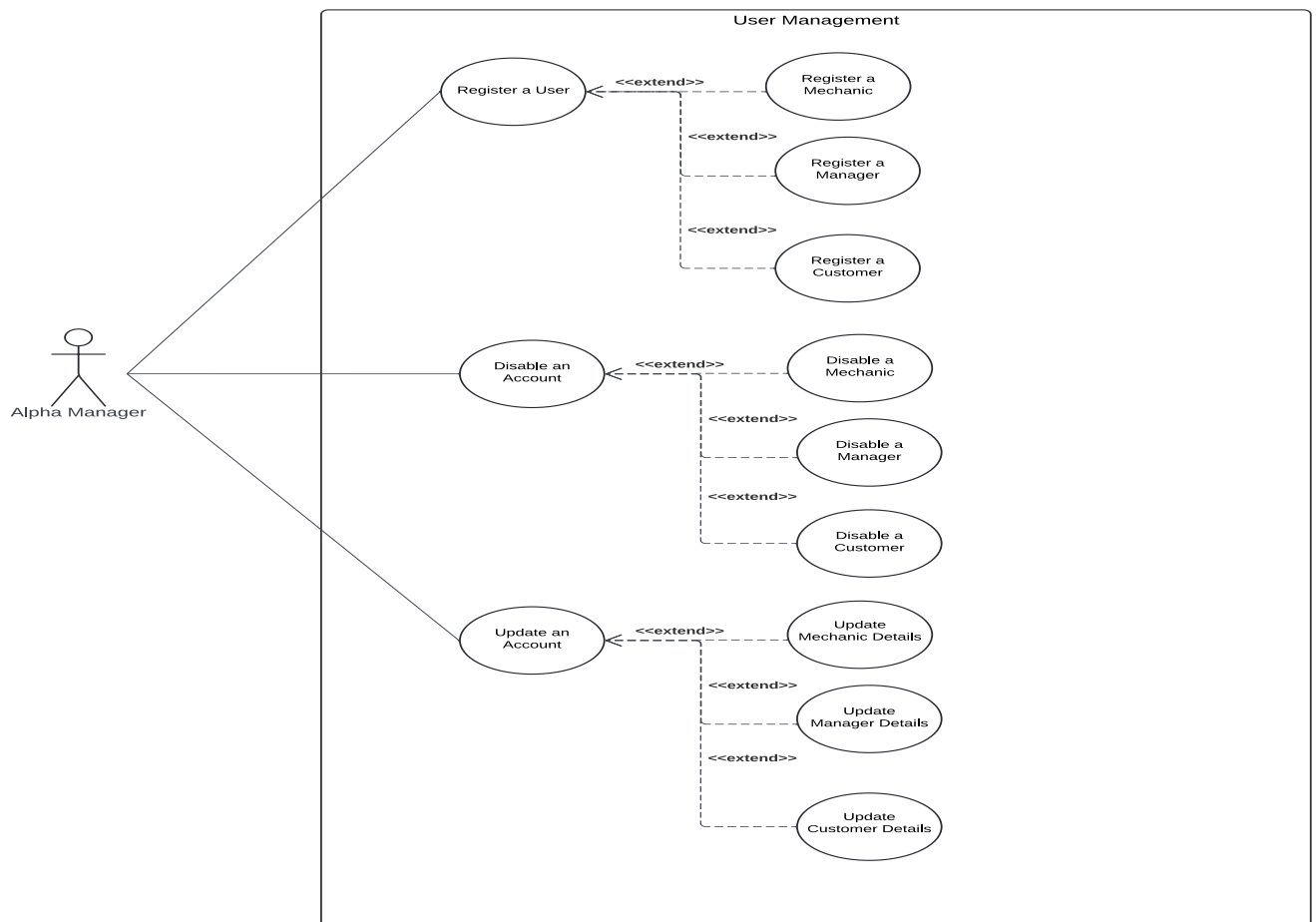


Figure 1: User Management Use Case Diagram

2. Service Request Management

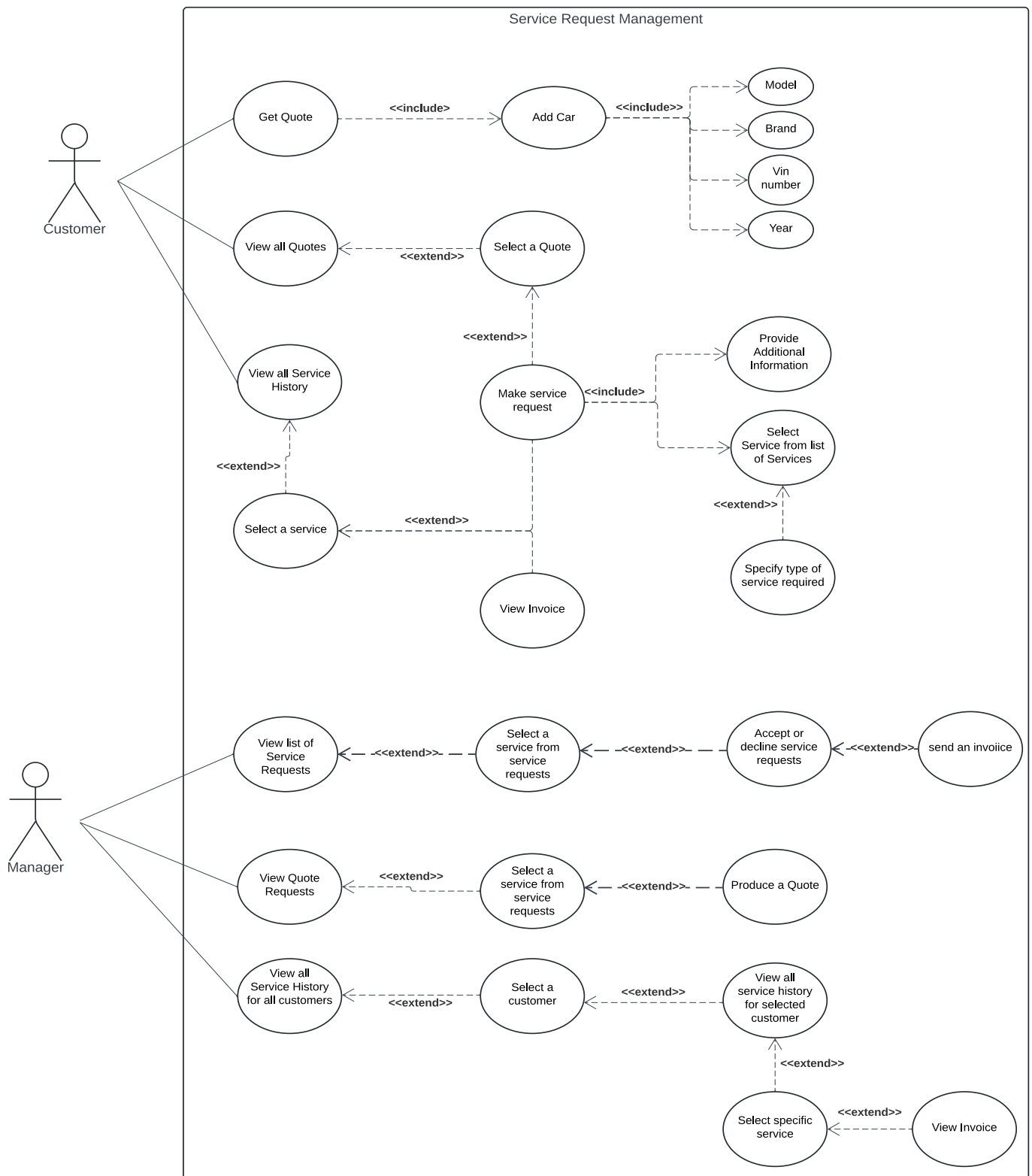


Figure 2: Service Request Management Use Case Diagram

3. Task Management:

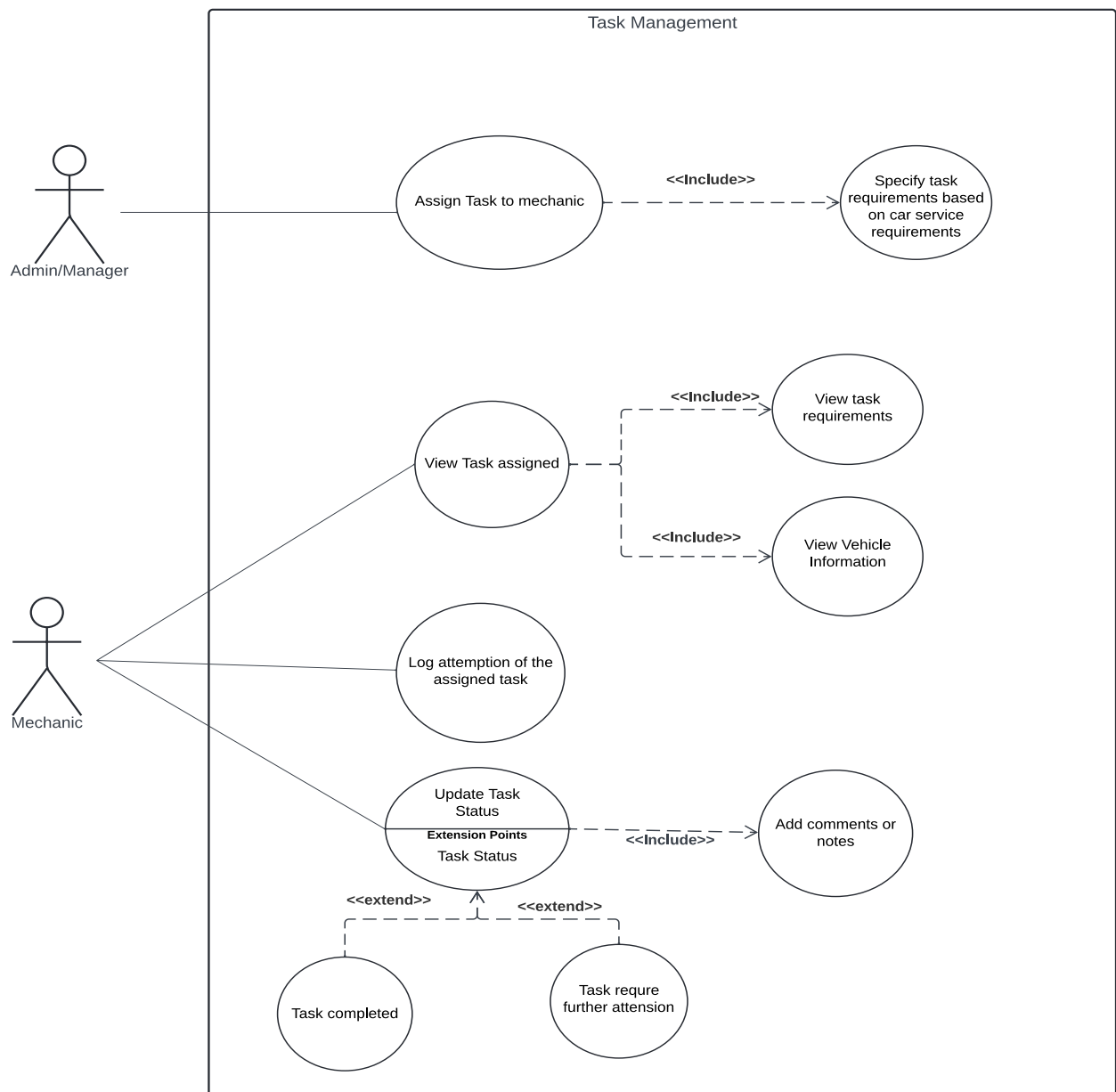


Figure 3: Task Management Use Case Diagram

4. Performance Monitoring:

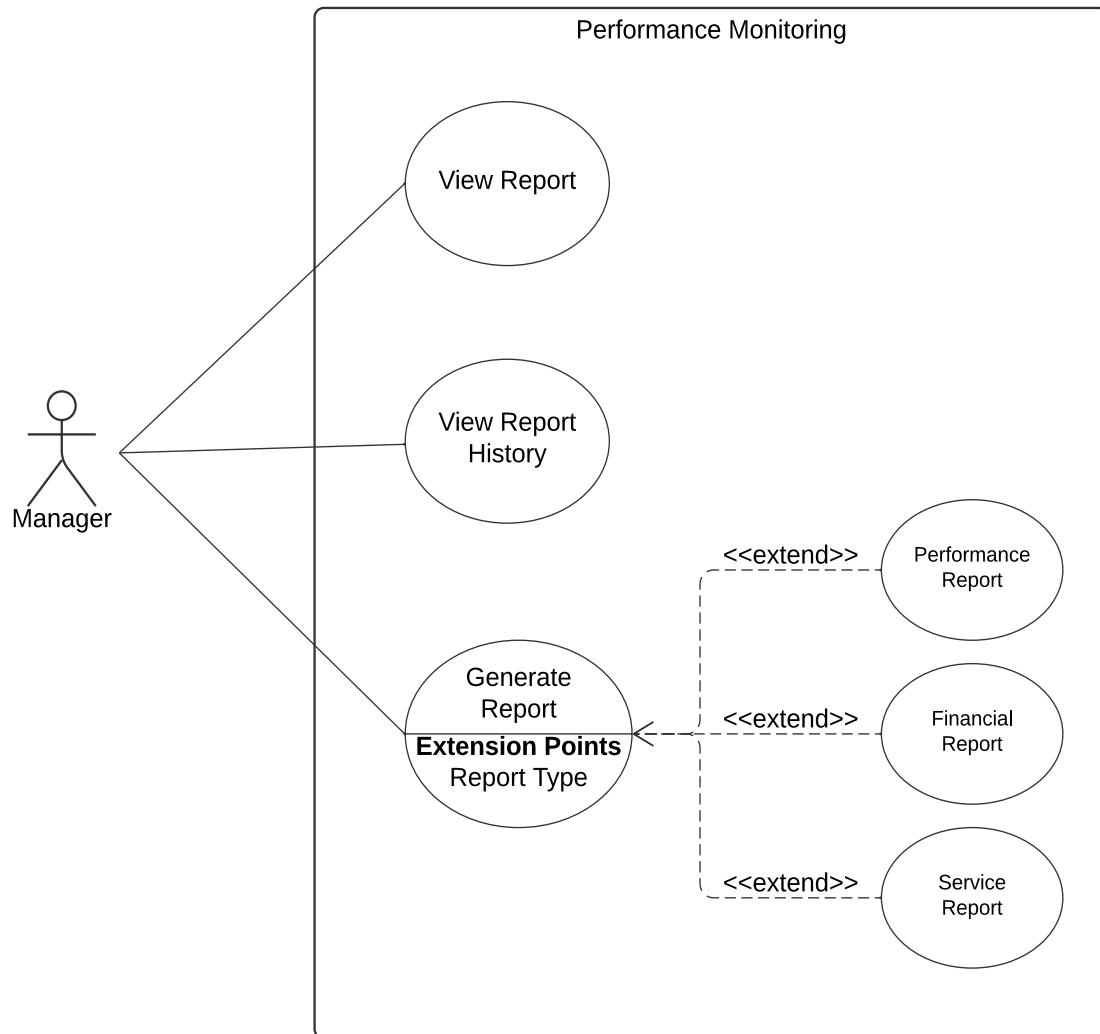


Figure 4: Performance Monitoring Use Case Diagram

5. Appointment Management:

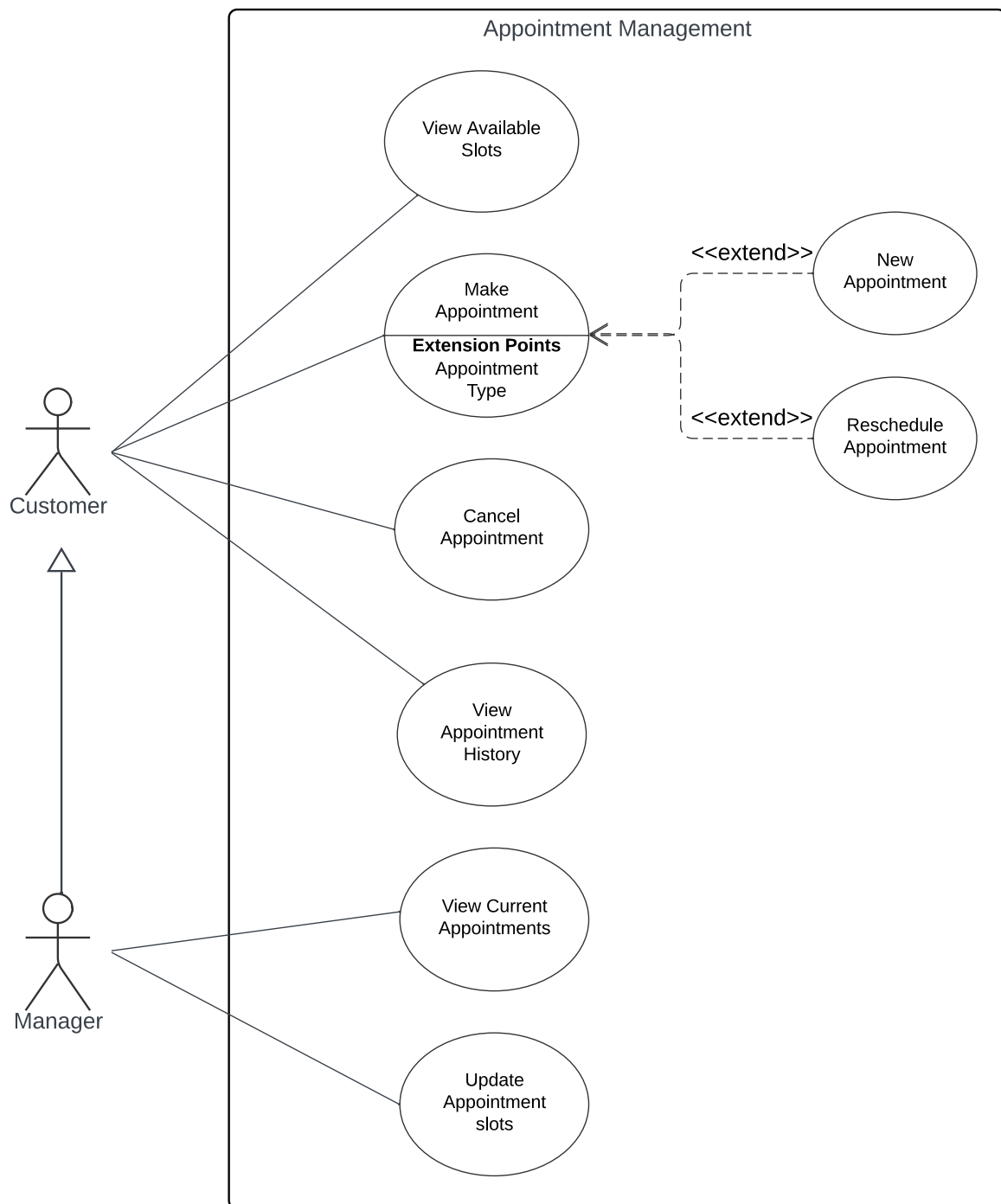


Figure 5: Appointment Management Use Case Diagram

6. Skills Sharing:

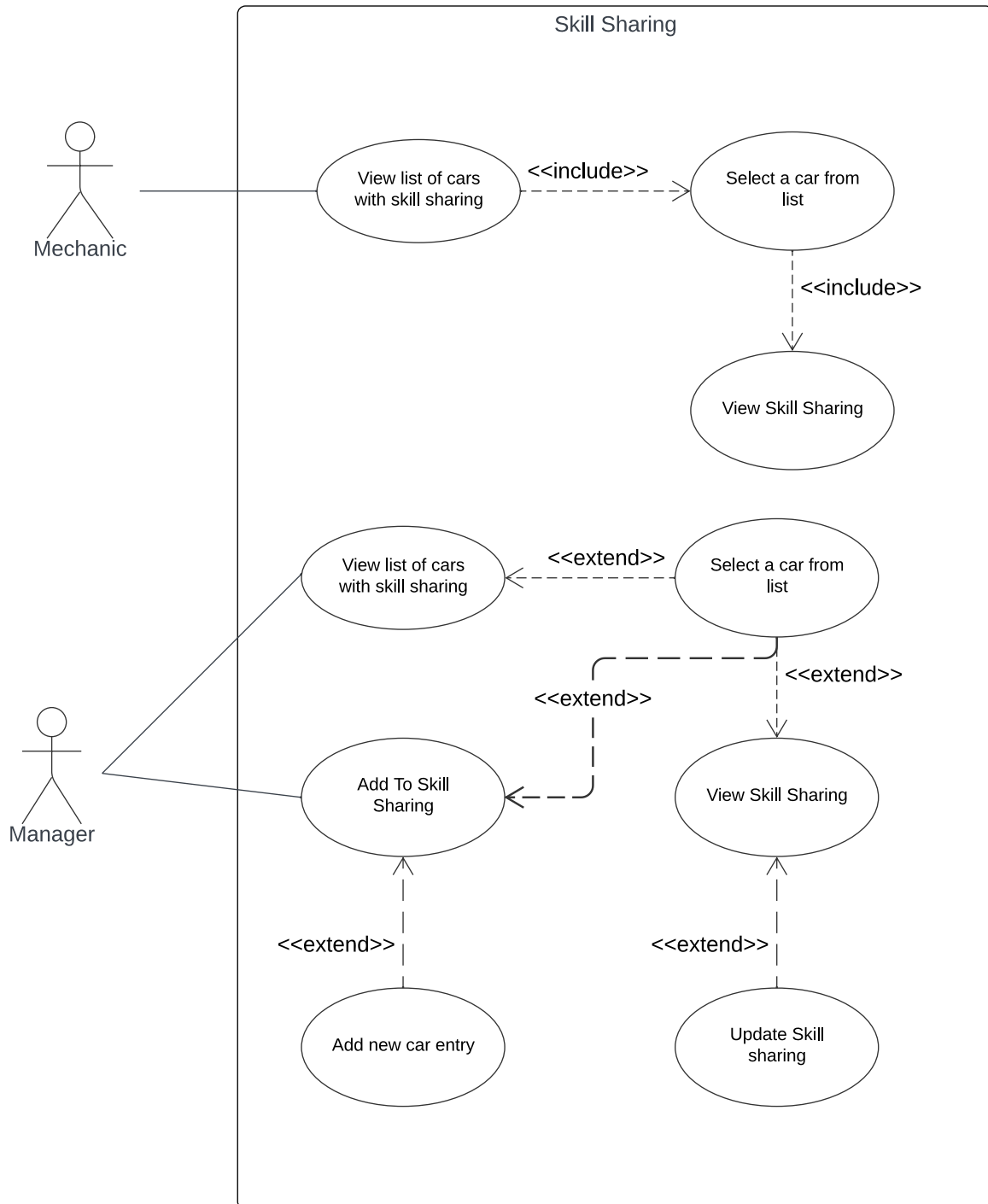


Figure 6: Skills Sharing Use Case Diagram

4. Feasibility and Risk Study

System feasibility

Our proposed system shares some similarities with workshop management systems such as “Workshop Software” developed by Xero. Workshop Software is an “Online workshop management software designed to effortlessly manage automotive, marine, truck and other types of workshops” (Workshop Software, 2016).

These similarities include online customer booking, quote and invoice generation, job management, and client and vehicle history management.

The workshop software has more features that are not on our proposed core functionalities. Therefore, considering that our system will not be developed on a large scale to be used by multiple workshops, we can conclude that overall, our proposed system is feasible in that respect.

Technical feasibility

The required hardware for the development of the system, is laptops with a minimum storage capacity of 250 GB of SSD, 8GB of RAM, and a multi-core processor for example Intel I5 or AMD Ryzen 5. This is to ensure that developers can run a database as well as backend and front-end applications all at the same time for development and test purposes.

All the development tools to be used are open-source or community versions therefore there will not be any costs related to the procurement of the software for the development of the system.

In terms of the operation of the system, the hardware requirements for the mobile application are a smart mobile Android phone that operates on at least Android 10 (API 29) for security purposes, (Stahie, 2022) earlier versions are becoming increasingly outdated and the last supported version is Android 10.

For the web application, the minimum required hardware is a laptop or a PC with at least 4 GB of RAM, storage of 250 GB, and a processor of at least Intel I3 or AMD Ryzen 3.

The procurement of the mentioned hardware and software will not be a problem therefore the system is technically feasible.

Economic feasibility

Using open-source and community versions of software, libraries, and APIs will ensure zero software procurement costs.

The hardware needed for the development of the system will be produced by us, as developers.

The hardware needed by the customer and mechanics for the operation of the system is a personal mobile phone.

The Manager will require a laptop or a desktop to operate the admin side of the system.

For the deployment and operation of the system, a cloud-based web hosting service will be required. Considering the scale of the business that this system is built for, an estimated fee for hosting the system ranges between R570.00 and R 950.00 per month (Bluehost, 2024).

Considering the projected RIO attributed to improved customer satisfaction and increased customer retention rate, we can conclude that the development and maintenance of the system are economically feasible for the client.

Operational feasibility

Technical Assistance

The workshop clients would also require a mobile phone, the mobile app would have a feature that will guide the user on how to operate the app.

Documentation for setting up, configuring, and operating the admin side of the system will be available on the web application, this will ensure that any user with the admin privilege can easily follow the steps on how to operate the system without any need for prior training.

Training and Education

The mechanics will need at least one week of training to integrate the system into their standard operating procedures.

The training could slow down the work rate in the workshop since it was proposed by the project sponsor that during the development of the system, we will have workshops with the mechanics to discuss how to make the features realistic and suit their needs, this should make the implementation support easier since the mechanics would already have an idea on how the system works. That will ensure that the system is operationally feasible.

5. References

Barnyard, A. W., n.d. *Auto Worx Barnyard*. [Online]

Available at: <http://autoworxbarnyard.co.za>

Bluehost, 2024. *Unmatched power, speed, & control with WordPress cloud hosting..* [Online]

Available at: <https://www.bluehost.com/hosting/cloud>

Buerkle Body Shop, 2014. *Auto Body Shop Vs Auto Repair Shop*. [Online]

Available at: <https://buerklebodyshop.com/auto-body-shop-vs-auto-repair-shop/>

Corrin, T., 2023. *Challenges Faced By Auto Repair Shops Without Software*. [Online]

Available at: <https://autoleap.com>

JMC Automotive Equipment, 2022. *5 Major Challenges Auto Repair Shops Face and How to Fix Them*. [Online]

Available at: <https://jmcautomotiveequipment.com/>

Leeming, D., 1974. The automobile repair industry. *Emerald Insight*.

Schmidt, G. & Withopf, J., 2014. Estimation of time-variant acoustic feedback paths in in-car communication systems. *In International Workshop on Acoustic Signal Enhancement*, pp. 104-108.

Stahie, S., 2022. *Coming Surge in Unsupported Android Devices Poses Growing Online Security Threat*. [Online]

Available at: <https://www.bitdefender.com/blog/hotforsecurity/coming-surge-in-unsupported-android-devices-poses-growing-online-security-threat/>

Workshop Software, 2016. *Workshop Software*. [Online]

Available at: <https://apps.xero.com/za/app/workshop-software#overview>